

BCA Python Programming Syllabus

Course: Python Programming

Section 1: Introduction to Python

- Lesson 1: History and Features of Python
- Lesson 2: Installing Python and IDE Setup
- Lesson 3: Variables and Data Types
- Lesson 4: Input and Output Operations

Section 2: Operators and Control Statements

- Lesson 1: Arithmetic, Relational and Logical Operators
- Lesson 2: if, if-else and nested if
- Lesson 3: for and while loops
- Lesson 4: break, continue and pass

Section 3: Strings and Collections

- Lesson 1: String Operations and Methods
- Lesson 2: Lists and List Methods
- Lesson 3: Tuples and Sets
- Lesson 4: Dictionaries and Dictionary Methods

Section 4: Functions and Modules

- Lesson 1: Defining and Calling Functions
- Lesson 2: Arguments and Return Values
- Lesson 3: Lambda Functions
- Lesson 4: Modules and Packages

Section 5: File Handling and Exception Handling

- Lesson 1: Reading and Writing Files
- Lesson 2: Working with CSV Files
- Lesson 3: try, except and finally
- Lesson 4: Raising Exceptions

Section 6: Object-Oriented Programming

- Lesson 1: Classes and Objects
- Lesson 2: Constructors
- Lesson 3: Inheritance
- Lesson 4: Polymorphism and Encapsulation

Section 7: Python Libraries for Data Science

- Lesson 1: Introduction to NumPy
- Lesson 2: Introduction to Pandas
- Lesson 3: Data Visualization with Matplotlib
- Lesson 4: Basic Data Analysis